

Integration and differentiation are, in some sense, inverse operations: $F(\theta) = \int_a^\theta f(t)dt \leftrightarrow F'(\theta) = f(\theta)$

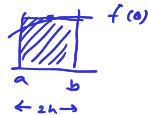
Fundamental theorem of calculus: if $F' = f$ then $F(b) - F(a) = \int_a^b f(t)dt$

$$\int_a^a f(t)dt = 0$$

recall: $\underbrace{F(\theta+h)}_b - \underbrace{F(\theta-h)}_a \approx \underbrace{F'(\theta)}_{f(\theta)} * 2h = \text{area under } f(\theta) \text{ over } 2h \text{ interval}$

$$f'(\theta) \approx \frac{F(\theta+h) - F(\theta-h)}{2h}$$

function $f(\theta)$	derivative $f'(\theta)$	integral $\int f(\theta)d\theta$
constant	0	constant * θ
θ^n	$n\theta^{n-1}$	$\frac{\theta^{n+1}}{n+1}$ (for $n \neq -1, \ln(\theta)$)
e^θ	e^θ	e^θ
$\ln(\theta)$	$\frac{1}{\theta}$	$\theta \ln(\theta) - \theta$
$\sin(\theta)$	$\cos(\theta)$	$-\cos(\theta)$
$\cos(\theta)$	$-\sin(\theta)$	$\sin(\theta)$



$$\frac{d(fg)}{d\theta} = f(\theta)g'(\theta) + f'(\theta)g(\theta) \rightarrow \int_a^b d(fg) = \int_a^b f g' d\theta + \int_a^b f' g d\theta$$

Integration by parts: $f g \Big|_a^b = \int_a^b \underline{f(\theta)} \underline{g'(\theta)} d\theta + \int_a^b f'(\theta) g(\theta) d\theta$

